

Pacific at Risk

The Climate Signal

Climate-Economic Vulnerability across Pacific Island Nations · 8 Datasets · 22 Countries

1990

2025

All

7.25

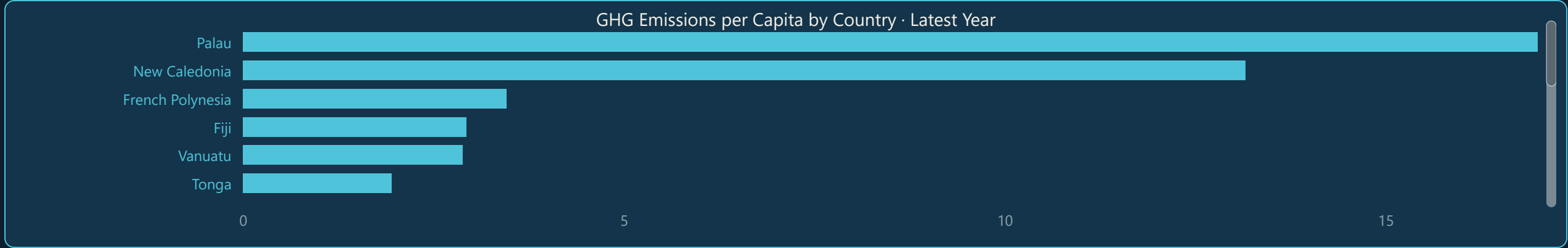
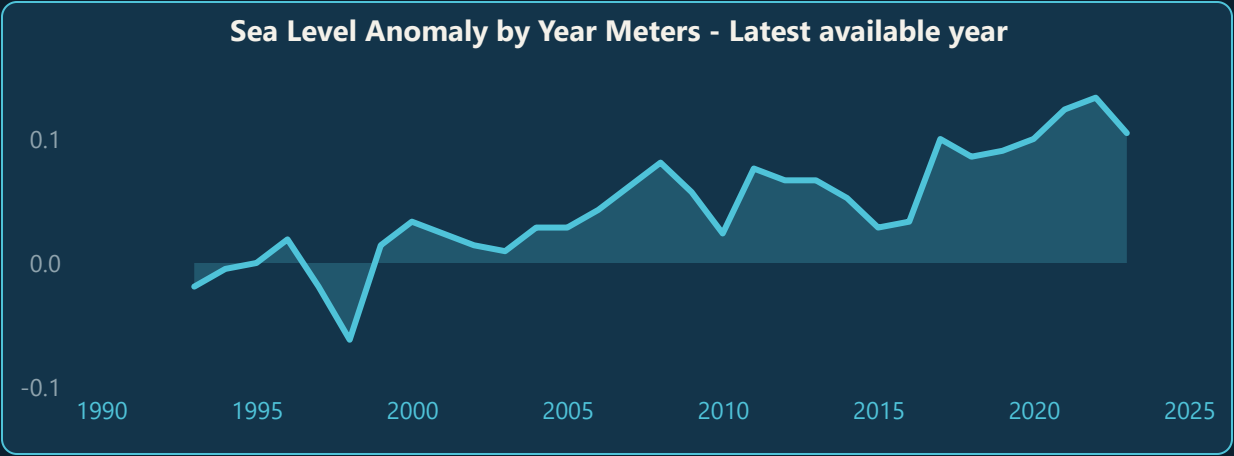
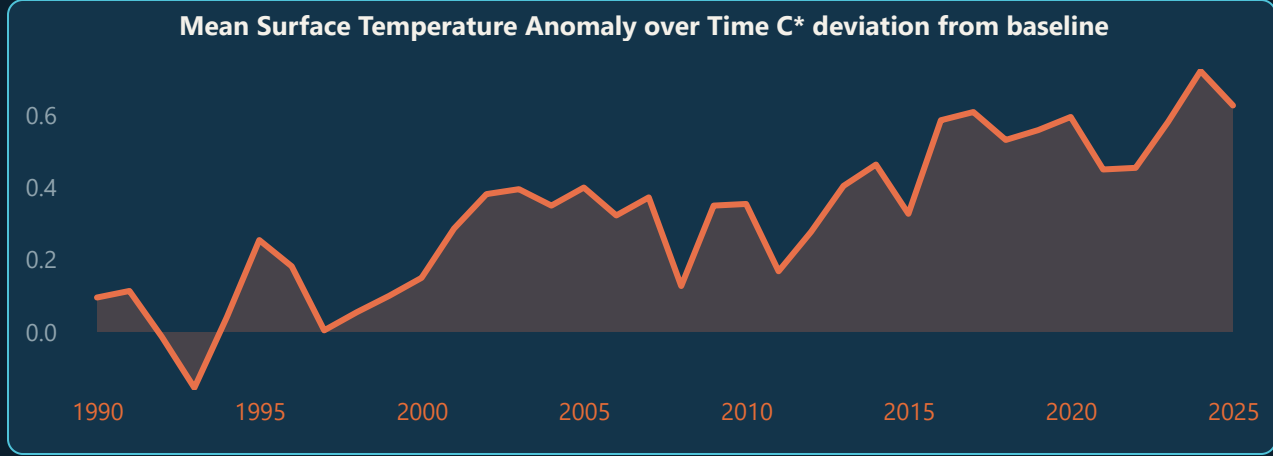
Avg GHG Latest Year

0.32

Avg Temp Anomaly (1990+)

0.10

Avg Sea Level Latest



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The Economic Cost

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2025

All

▼

Avg Loss per Person Affected (USD)

959M

Total Persons Affected by Disasters

2.59M

Avg Loss per Person Affected (USD)

370.45

Direct Disaster Economic Loss by Country

Country	Loss (bn USD)
Fiji	0.55
Vanuatu	0.18
Tonga	0.08
French Polynesia	0.06
Federated States of Micronesia	0.05

Persons Affected by Disasters over Time

▲ Cyclone Winston 2016

Year	Persons Affected (M)
2005	0.01
2006	0.05
2007	0.02
2008	0.01
2009	0.01
2010	0.01
2011	0.01
2012	0.02
2013	0.01
2014	0.01
2015	0.15
2016	0.70
2017	0.02
2018	0.45
2019	0.10
2020	0.55
2021	0.08
2022	0.05
2023	0.01

GHG Emissions vs Disaster Economic Loss by Country

Each bubble = one country · Higher emissions correlate with higher economic loss

▲ Palau excluded (86t GHG) & New Caledonia excluded as extreme outlier

Country	GHG Emissions (Gt)	Economic Loss (bn USD)
Fiji	2.8	0.55
Vanuatu	2.8	0.15
Samoa	1.8	0.05
Tonga	1.9	0.05
Papua New Guinea	1.4	0.05
Solomon Islands	0.9	0.05
Kiribati	0.6	0.05
Federated States of Micronesia	0.4	0.05
Marshall Islands	0.1	0.05
French Polynesia	3.2	0.02

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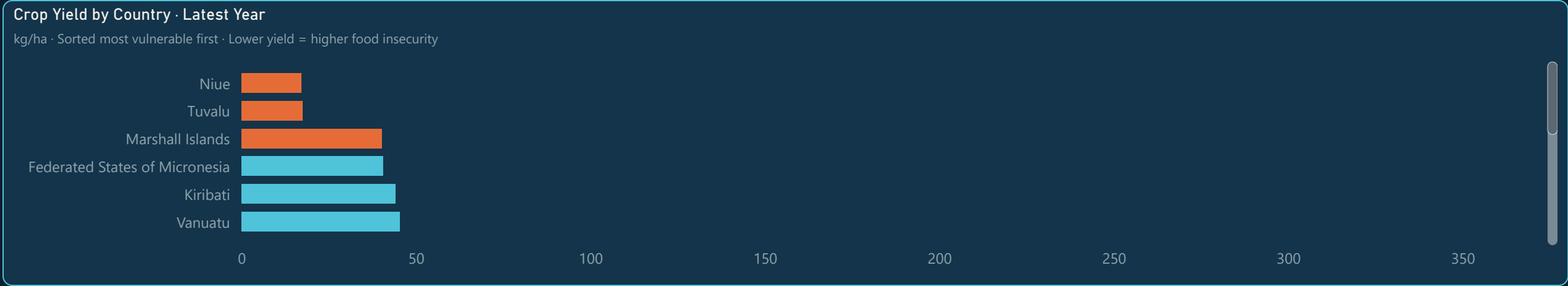
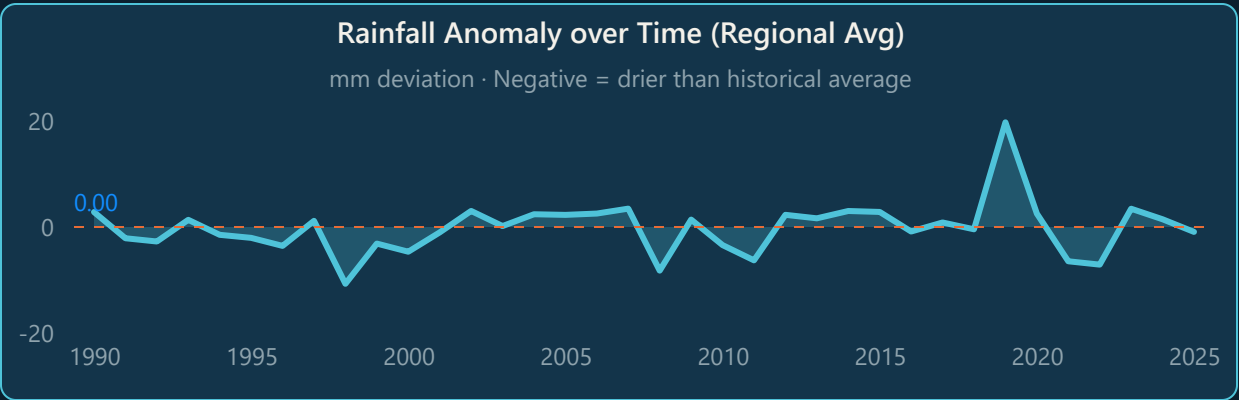
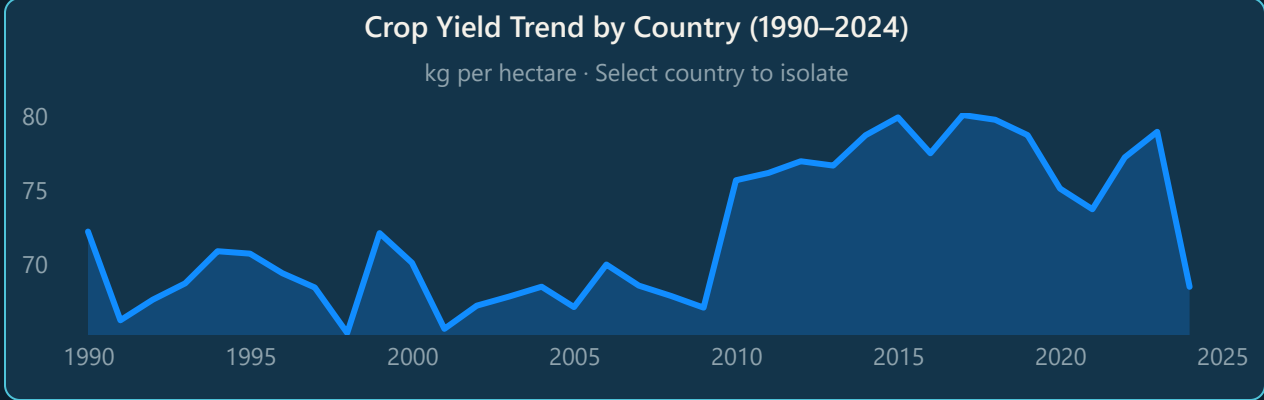
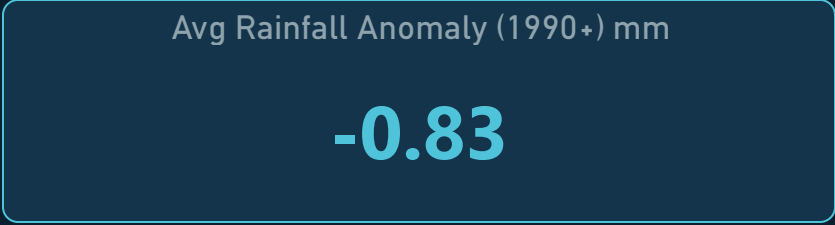
Food & Livelihoods

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1990

2025

All



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Vulnerability Index

19902025

All

Nations at High Risk

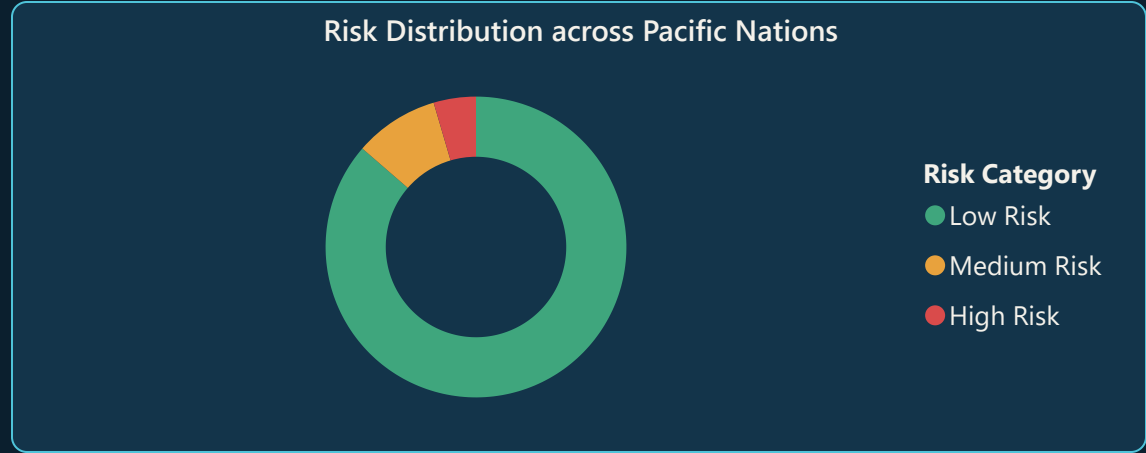
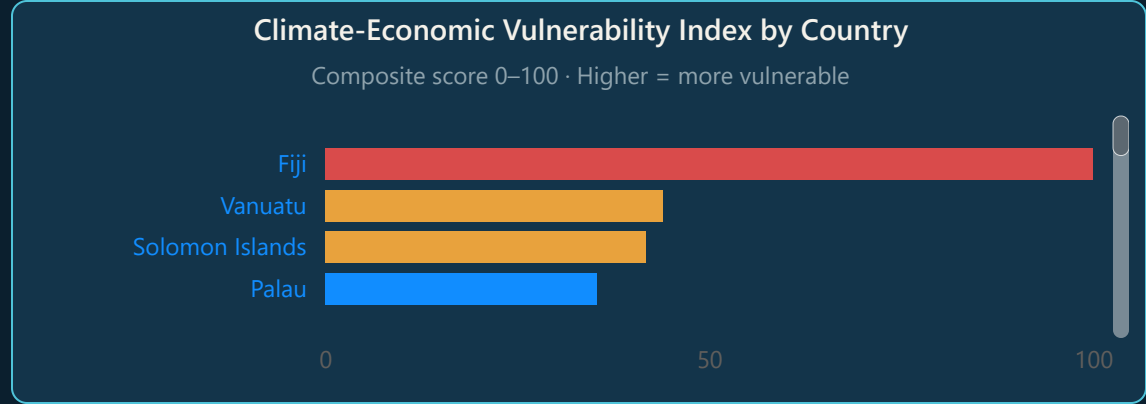
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Avg Vulnerability Score (0–100)

27.33

Avg Disaster Loss

25M



Vulnerability Score Components by Country					
Country	GHG Score	Disaster Score	Persons Score	Crop Score	Vulnerability Index
Wallis and Futuna	0.00	0.00	0.00	100.79	20.16
Vanuatu	1.38	40.43	43.22	93.76	43.98
Tuvalu	0.17	0.00	1.74	98.07	20.08
Tonga	0.93	9.61	14.19	89.44	24.99
Tokelau	0.00	0.00	0.00	100.79	20.16
Solomon Islands	0.45	5.25	88.76	88.74	41.87
Samoa	0.90	5.89	4.64	92.89	21.98
Pitcairn	0.00	0.00	0.00	100.79	20.16
Papua New Guinea	0.56	3.72	15.67	89.07	23.15
Palau	62.95	3.25	5.92	100.79	35.37
Northern Mariana Islands	0.05	0.00	0.00	100.79	20.17
Niue	0.00	0.00	0.00	98.13	19.63
New Caledonia	6.28	5.42	0.00	91.98	21.55
Nauru	0.05	0.00	1.04	90.54	18.38
Marshall Islands	0.05	1.90	28.46	94.57	26.70
Kiribati	0.31	0.07	0.47	93.96	19.00
Guam	0.06	0.00	0.00	100.79	20.17
French Polynesia	1.65	8.07	0.00	91.21	21.40
Fiji	1.40	164.92	195.83	43.33	100.00
Fiji (excl. GHG & Crop)	0.10	7.00	7.04	94.51	22.20